

Project 2: Crash Data

STAT 133, Spring 2026

About

The purpose of this assignment is to give you the opportunity to practice data manipulation with tidyverse tools on a real data set. Likewise, this project will allow you to put in practice your quarto skills to create [Revealjs](#) slides.

General Instructions

- Use a Quarto Document (.qmd file) for your Project 2.
- Save this file as `project2.qmd`.
- You will have to Render your `qmd` file into revealjs slides. In other words, your output format in the YAML header should be `format: revealjs`. This was covered in lab-4.
- Like in project-1, assume that your published slides will be seen by a potential employer, boss, client/customer, admissions committee, or something like that. In other words, make sure that the content of your project is “understandable”: well organized, self descriptive, easy to follow, etc.
- To receive full credit, please submit the following to bCourses (proj2 assignment):
 - `qmd` file
 - (URL) link to your published document
- If possible, it would be nice if you can publish your `qmd` file to your **QuartoPub** account.

Content Notice

The content of this assignment includes crash data in California. This content was obtained through UC Berkeley’s *Safe Transportation Research and Education Center* (SafeTREC).

The decision to include such material is not taken lightly, and we’re doing so in order to achieve the learning goals for this course: to apply the analytical and coding skills utilizing real-world data sets.

We want to do our best to make any conversations we have about this content in class feel safe for you. If you have thoughts to share with us about this, please do send us a note through Ed or email.

1 Data

The data for this project—obtained through UC Berkeley SafeTREC—has to do with the **Statewide Integrated Traffic Records System** (SWITRS) which is a data set collected and maintained by the California Highway Patrol (CHP).

Files

There are 2+1 files associated with the data behind this project:

- 1) `crashes_2023.csv`: Data file in CSV format
- 2) `Codebook-Crash-Data-2023.pdf`: Documentation (i.e. metadata) about the data columns in the CSV file.

Both files (CSV & PDF) are available in bCourses (see **Files/projects/project2**):

<https://bcourses.berkeley.edu/courses/1551809/files/folder/projects/project2>

Importing CSV file in a qmd file

Assuming that the CSV is in your working directory, and that you've loaded the package "tidyverse", you can quickly import the data with the function `read_csv()`. Here's a code snippet for how to do this:

```
library(tidyverse)
dat <- read_csv("crashes_2023.csv")
```

Sample Slides

In case of need, you will also be able to find some sample slides: source file `sample-slides.qmd` and rendered file `sample-slides.html`. See folder **Files/projects/project2**.

2 Prompts

💡 Specs

The main goal in this project is to obtain summary statistics that allow the viewer to have a general understanding of a handful of aspects about crashes in California during 2023.

- You will have to deliver your report in the form of Revealjs slides.
- Since we haven't spent much time around data visualization, graphics are welcome but not required.
- Take advantage of **code folding** as well as **inline code** (covered in lab-4).
- You will have to address the following prompts by writing "dplyr" commands.

2.1 Title Slide and About Slide

Make sure your presentation includes a Title slide, and also an "About" or "Agenda" slide (embedding images is encouraged!)

2.2 Overall Statistics

Write *dplyr* commands to obtain the following summaries (1 slide). Use code-folding and inline code.

- Total number of crashes.
- Total number of injured victims.
- Total number of fatal victims.

2.3 Distribution Tables

Write *dplyr* commands to obtain the following (1 slide per table; use code-folding):

- Table with distribution (counts and percentages) of crashes per **month**.
- Table with distribution (counts and percentages) of crashes per **day of week**.
- Table with distribution (counts and percentages) of crashes per **hour**.

2.4 Ranking Tables

Write *dplyr* commands to obtain the following (1 slide per table, use code-folding):

- Table of Top-10 **PCF Violations** (from most to least common)
- Table that ranks the **Types of Collision** (from most to least common)
- Table of Top-8 **Party At Fault** (from most to least common)

2.5 Crashes involving pedestrians

Write *dplyr* commands to obtain the following counts (1 slide). Use code-folding and inline code.

- Number of accidents involving pedestrians
- Number of injured pedestrians
- Number of killed pedestrians

Write *dplyr* commands to obtain the following (1 slide per table, use code-folding):

- Table with Top-5 **PCF Violations** for pedestrian accidents
- Table with Top-5 **Types of Collision** for pedestrian accidents
- Table with Top-5 **Party at Fault** for pedestrian accidents

2.6 Crashes involving bicyclists

Write *dplyr* commands to obtain the following counts (1 slide). Use code-folding and inline code.

- Number of accidents involving bicyclists
- Number of injured bicyclists
- Number of killed bicyclists

Write *dplyr* commands to obtain the following (1 slide per table, use code-folding):

- Table with Top-5 **PCF Violations** for pedestrian accidents
- Table with Top-5 **Types of Collision** for pedestrian accidents
- Table with Top-5 **Party at Fault** for pedestrian accidents

2.7 Choose your own exploration

Come up with an exploratory research question that you would like to answer by analyzing `crashes_2023.csv`. Use 1 or 2 slides for this task.